

A counterexample to the convex-block difference-basis problem

arXiv:1711.01340, final Problem 7

June 14, 2026

Source question

The source paper asks the following final problem.

Let X be a Banach space with a convex block homogeneous basis (e_i) . Define

$$d_1 = e_1, \quad d_{i+1} = e_{i+1} - e_i \quad (i \geq 1).$$

Is X , endowed with the basis (d_i) , submultiplicative?

Problem 5. Find a Banach space X with an unconditional basis $(x_i)_i$ that is not equivalent to the unit vector basis of c_0 so that there exist a Banach space the Calkin algebra of which is isomorphic as a Banach algebra to the unitization of X .

Remark 8.9. The unitization of James space $\mathbb{R}e_\omega \oplus J$ may be viewed as the space of all scalar sequences of bounded quadratic variation and similarly for every Banach space X with a subsymmetric sequence the space $\mathbb{R}e_\omega \oplus J(X)$ is the space of all scalar sequences of bounded X -variation. One may consider for any such X and any linearly ordered set I the space $V_X(I)$ of all functions $f : I \rightarrow \mathbb{R}$ of bounded X -variation. The norm on such a space is submultiplicative. The spaces $V_X[0, 1]$ were introduced in [AMP]. By very carefully combining the method of the present paper with the method from [MPZ] it is conceivable that one may obtain for every countable well ordered set I a Banach space the Calkin algebra of which is isomorphic as a Banach algebra to $V_X(I)$.

Problem 6. Let X be a Banach space with a subsymmetric basis. For what ordered sets I does there exist a Banach space the Calkin algebra of which is isomorphic as a Banach algebra to $V_X(I)$?

Remark 8.10. In [AMS] the notion of a convex block homogeneous sequence is introduced, i.e., a sequence that is equivalent to its convex block sequences. Known examples of such sequences are constructed using a space X with a subsymmetric basis $(x_i)_i$. The bases $(e_i)_i$ of $J(X)$ and $(v_i)_i$ of $\mathcal{J}_*(X)$ are both convex block homogenous. In either case, the difference basis is submultiplicative.

Problem 7. Let X be a Banach space with a convex block homogeneous basis $(e_i)_i$ and set $d_1 = e_1$ and for $i \in \mathbb{N}$ $d_{i+1} = e_{i+1} - e_i$. Is X endowed with $(d_i)_i$ submultiplicative?

Counterexample

Theorem. Problem 7 of arXiv:1711.01340 has a negative answer as printed. In ℓ_2 , the usual unit vector basis is convex block homogeneous, but the difference sequence

$$d_1 = e_1, \quad d_{i+1} = e_{i+1} - e_i$$

does not satisfy any uniform submultiplicative estimate for finite coordinate vectors.

Proof. The usual unit vector basis of ℓ_2 is convex block homogeneous: every normalized block sequence is orthonormal, hence isometric to the original basis. In particular, every normalized convex block sequence has the same property.

For a finitely supported scalar sequence $a = (a_i)$, with $a_i = 0$ after the support, one has

$$\left\| \sum_i a_i d_i \right\|_2^2 = \sum_i |a_i - a_{i+1}|^2.$$

Fix $N \geq 2$ and set

$$a_i = \begin{cases} i/\sqrt{N}, & 1 \leq i \leq N, \\ (2N - i)/\sqrt{N}, & N < i \leq 2N, \\ 0, & i > 2N. \end{cases}$$

Then

$$\left\| \sum_i a_i d_i \right\|_2^2 = \frac{2N - 1}{N} < 2.$$

On the other hand, applying the same coordinate vector in both factors would force control of

$$\left\| \sum_i a_i^2 d_i \right\|_2^2 = \sum_i |a_i^2 - a_{i+1}^2|^2.$$

Already the increasing side gives

$$\left\| \sum_i a_i^2 d_i \right\|_2^2 \geq \frac{1}{N^2} \sum_{i=1}^{N-1} ((i+1)^2 - i^2)^2 = \frac{1}{N^2} \sum_{i=1}^{N-1} (2i+1)^2,$$

which tends to infinity as $N \rightarrow \infty$.

Thus no constant C can satisfy

$$\left\| \sum_i a_i b_i d_i \right\|_2 \leq C \left\| \sum_i a_i d_i \right\|_2 \left\| \sum_i b_i d_i \right\|_2$$

for all finitely supported coefficient vectors a, b , since taking $b = a$ gives a bounded right-hand side and an unbounded left-hand side. $\square$

Remark. *The argument answers the literal printed formulation. If the intended question silently assumes that (d_i) is already a Schauder basis, or imposes another additional hypothesis excluding the classical Hilbert basis, this packet should be read as a wording counterexample rather than as a solution of that strengthened variant.*

Verification and provenance

The finite-dimensional calculation is checked independently in the bundled verification script. The packet README records the duplicate and bounded novelty checks performed on 2026-06-14. Local run indexes contained no prior packet for this exact convex-block difference-basis problem; web searches found only the source paper context, not a later answer.

Human Review: convex-block homogeneous misreading

Packet identifier: 1711.01340_ell2_difference_basis_not_submultiplicative

Original automatic proof: `solution_packet.pdf`

Checked date: 14 June 2026

Status: Manually checked.

Verdict: Rejected as a solution to the intended problem. This is a miss.

Human Comments

The proposed solution in `solution_packet.pdf` should not be accepted as a counterexample to Problem 7 of Motakis–Puglisi–Tolias. The finite calculation with the difference sequence in ℓ_2 is correct, but the example does not satisfy the hypothesis of the problem.

The packet claims that the usual unit vector basis of ℓ_2 is convex block homogeneous because every normalized block sequence is orthonormal. This uses the wrong notion. In the context cited by the source paper, Argyros–Motakis–Sari introduce a convex block homogeneous sequence as one which is equivalent to its convex block sequences. The convex blocks are not renormalized.

For the unit vector basis of ℓ_2 , take successive finite sets B_k with $|B_k| \rightarrow \infty$, and define

$$y_k = \frac{1}{|B_k|} \sum_{i \in B_k} e_i.$$

Then (y_k) is a convex block sequence of the unit vector basis, but

$$\|y_k\|_2 = |B_k|^{-1/2} \longrightarrow 0.$$

Therefore (y_k) is not even seminormalized and cannot be equivalent to the normalized unit vector basis of ℓ_2 . Thus the proposed example fails the defining hypothesis “convex block homogeneous basis”.

This also fits the surrounding context of the source paper. Remark 8.10 mentions the known convex block homogeneous examples arising from $J(X)$ and $J^*(X)$, and then says that a positive answer to Problem 7 would imply a statement about difference bases of conditional spreading sequences. The question is not aimed at the Hilbert unit vector basis under a normalized-block interpretation.

There is a secondary warning as well: in ℓ_2 , the sequence

$$d_1 = e_1, \quad d_{i+1} = e_{i+1} - e_i$$

is not a Schauder basis of ℓ_2 . The inverse coordinate map involves partial summation and is unbounded. This reinforces that the packet is testing a coefficient norm outside the intended setting.

Short Version

This is a miss. The packet treats “convex block homogeneous” as equivalence to normalized block sequences, but the source context uses equivalence to convex block sequences themselves. The ℓ_2 unit vector basis fails that hypothesis by taking long equal-average convex blocks.

Review Outcome

Reject this candidate as a proposed counterexample to Problem 7. It may be archived as a warning that automatic extraction must preserve the precise definition of “convex block homogeneous” and the surrounding conditional spreading-sequence context.